PETERSON ELEMENTS AND THE STEENROD ACTION ON THE DICKSON–MÙI ALGEBRAS

ABSTRACT. We describe the algorithm used in `yacop/modules/dickson.py` to compute the action of the mod p Steenrod algebra on the Dickson algebras D_k (and on their Mùi extensions). The algorithm rests on a family of invariants, the *Peterson elements* $\widetilde{\omega}_n$, and on a closed formula $P^R\widetilde{\omega}_n = a(R,n)\,\widetilde{\omega}_{n+d(R)}$ for the action of a Milnor basis element on them. We define the Peterson elements, give an explicit monomial formula for them which generalises a formula of Hung and Peterson, and prove that the Dickson invariants are Peterson elements. We extract the action formula from the code. For odd p it holds with the normalisation $\widetilde{\omega}_n = (-1)^n\omega_n$, not with the one stated in the code's documentation. We prove the formula for all Peterson elements, all k and all Milnor basis elements. In particular the algorithm is correct. We also prove the code's formula for the conjugate action. Finally we explain the auxiliary class `DicksonDualSteenrodAlgebra`.

1. INTRODUCTION

The program `yacop` computes the action of the Steenrod algebra $\mathcal{A}$ on the Dickson algebras $D_k = H^*(BV_k;\mathbb{F}_p)^{\mathrm{GL}_k}$, where $V_k = \mathbb{F}_p^k$. Here D_k is viewed as an algebra of invariants in the polynomial part of $H^*(BV_k)$. Since D_k is a polynomial algebra, the Cartan formula reduces the problem to the action on the generators. The code handles the generators by identifying each of them with a *Peterson element* and then using a closed formula for the Milnor basis action on Peterson elements. This note explains that formula and how much of it can be proved.

Summary of results. Let $R = (r_1, r_2, \dots)$ index the Milnor basis element P^R (written $Sq(R)$ when $p = 2$). Put

$$|R| = \sum_j r_j, \qquad N(R) = \sum_j p^j r_j, \qquad d(R) = \sum_j r_j\,\frac{p^j-1}{p-1},$$

so that $(p-1)d(R) = N(R) - |R|$. For $n \geq 0$ define the coefficient

$$a(R,n) \;=\; \binom{(p-1)(n+d(R))}{N(R)}\binom{N(R)}{pr_1,\, p^2r_2,\, p^3r_3,\, \dots} \in \mathbb{F}_p, \tag{1}$$

where the second factor is a multinomial coefficient. The code (functions `__action_exponents` and `__action_exponents2` in `dickson.py`) uses

$$P^R\,\widetilde{\omega}_n \;=\; a(R,n)\,\widetilde{\omega}_{n+d(R)}. \tag{$*$}$$

- Section 3 defines Peterson elements and proves the monomial formula and the identification of the Dickson invariants (Proposition 3.2, Theorem 3.4).
- Section 4 describes the algorithm. The code's decomposition of Peterson elements into Dickson monomials is a Lagrange inversion formula (Proposition 4.2).
- Section 5 proves ($*$) at the Dickson generators (Theorem 5.9), which shows that the algorithm is correct (Corollary 5.10). It then proves ($*$) in all degrees (Theorem 5.11), and the formula used by the code for the conjugate action (Theorem 5.14). Table 1 lists independent computer checks.

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- Section 6 explains how the code obtains the action on all of D_k, Section 7 derives the action on the Mùi generators, and Section 8 explains `DicksonDualSteenrodAlgebra`.

Remark 1.1 (A sign). The code documents its Dickson generators as the Peterson elements $\omega_n = (x_1 \cdots x_k)\,\chi(P^n)\big((x_1 \cdots x_k)^{-1}\big)$. For odd p, formula $(*)$ is *false* for the ω_n; for example $P^1\omega_1 = -\omega_2$ in $H^*(B\mathbb{Z}/3)$, while $(*)$ predicts $+\omega_2$. Formula $(*)$ is correct for $\widetilde{\omega}_n = (-1)^n\omega_n$ (Theorem 5.11, together with Corollary 5.10 for the comparison of $\widetilde{\omega}_n$ and Ω_n). Both the decomposition formula and the Cartan formula are insensitive to this sign change: n is additive under products, so a Dickson monomial in the $\widetilde{\omega}$'s picks up the same sign $(-1)^n$ as $\widetilde{\omega}_n$ itself. So the decomposition formula is the same for ω and $\widetilde{\omega}$, and the code's generators are consistent with $\widetilde{\omega}_{n_j}$. With that reading the module computed by the code is the correct one. Only the identification "generator $= \omega_n$" in the documentation should read "generator $= \widetilde{\omega}_n$". For $p = 2$ there is no difference.

2. Recollections

2.1. **Steenrod operations.** Write $H = \mathbb{F}_p[x_1, \dots, x_k]$, with $|x_i| = 1$ for $p = 2$ and $|x_i| = 2$ for p odd. This is the polynomial part of $H^*(BV_k)$. (For odd p, $H^*(BV_k) = E(y_1, \dots, y_k) \otimes H$ with $|y_i| = 1$, and the Bockstein maps $y_i \mapsto x_i$.) We use the Milnor basis $\{P^R\}$ of the subalgebra of $\mathcal{A}$ generated by the reduced powers. Let $t = (t_1, t_2, \dots)$ be indeterminates and put $t_0 = 1$. The *total Milnor operation*

$$\Lambda_t(y) = \sum_R t^R P^R(y), \qquad t^R = \prod_j t_j^{r_j},$$

is a ring homomorphism $H \to H[t]$, extended $\mathbb{F}_p[t]$-linearly. It is determined by

$$\Lambda_t(x) = T(x) := \sum_{j \geq 0} t_j\, x^{p^j} \qquad (x \in H^{|x_1|}).$$

Since the elements Λ_t are group-like, the conjugate operation $\Lambda'_t(y) = \sum_R t^R \chi(P^R)(y)$ is the inverse of Λ_t. On a degree-one (resp. degree-two) class it is $\Lambda'_t(x) = T^{-1}(x)$, where T^{-1} is the compositional inverse of the additive series T. (Indeed $\Lambda_t\big(\sum_j c_j x^{p^j}\big) = \sum_j c_j T(x)^{p^j}$ for scalars $c_j \in \mathbb{F}_p[t]$.)

2.2. **Dickson invariants.** Let $V \subset H$ be the $\mathbb{F}_p$-span of $x_1, \dots, x_k$. The Dickson polynomial

$$f_V(X) \;=\; \prod_{v \in V} (X - v) \;=\; \sum_{i=0}^{k} c_{k,i}\, X^{p^i}, \qquad c_{k,k} = 1,$$

is additive in X. Its coefficients $c_{k,i}$ are the (signed) Dickson invariants. In the classical notation $c_{k,i} = (-1)^{k-i} Q_{k,i}$, and $|c_{k,i}| = (p^k - p^i)|x_1|$. Dickson's theorem says $D_k = H^{\mathrm{GL}_k} = \mathbb{F}_p[c_{k,0}, \dots, c_{k,k-1}]$ [1]. Two facts about additive polynomials will be used repeatedly:

(D1) An additive polynomial with $\mathbb{F}_p$-coefficients is $\mathbb{F}_p$-linear. So it vanishes on V as soon as it vanishes on $x_1, \dots, x_k$.

(D2) Over a domain, a monic polynomial of degree p^k that vanishes on the p^k distinct elements of V equals f_V.

Since Λ_t is a ring homomorphism,

$$\Lambda_t(f_V)(X) = \prod_{v \in V} \big(X - T(v)\big) = f_{T(V)}(X). \tag{2}$$

2.3. **The Mùi extension.** For odd p, Mùi [8] computed the full ring of invariants $H^*(BV_k)^{\mathrm{GL}_k}$. It is generated over D_k by exterior-type classes $R_{k,i}$ $(0 \le i < k)$ of degree $2(p^k-1)-2p^i+1$, with $R_{k,i}^2 = 0$. These are built from Moore-type determinants with one row of y's; see also [6]. The `DicksonAlgebra` class with `generic=True` adds odd generators τ_i of degree $1-2p^i$ to D_k. These are the invariants $R_{k,i}/Q_{k,0}$ (up to sign). Their action is derived in Section 7. Apart from that section, this note concentrates on the polynomial part D_k.

3. Peterson elements

Let $S(z) = \sum_{j\ge 0} z^{p^j}$. This is the compositional inverse of the additive polynomial $z \mapsto z - z^p$, i.e.

$$S(z) - S(z)^p = z. \tag{3}$$

Definition 3.1. Let $e = x_1 \cdots x_k$. For $n \ge 0$ the *Peterson element* $\omega_n \in H$ is

$$\omega_n = e \cdot \chi(P^n)(e^{-1}).$$

Here e^{-1} lives in the localisation $H[e^{-1}]$, on which $\mathcal{A}$ acts by the Cartan formula. The *normalised* Peterson element is $\widetilde{\omega}_n = (-1)^n \omega_n$. It has degree $(p-1)n|x_1|$. (Code: `PetersonPolynomials.omega`, with `_omega_safe` implementing the definition.)

For $p = 2$ these are the elements studied by Hung and Peterson [5, §2].

Proposition 3.2. (1) *In the completion of H,*

$$\sum_{n\ge 0} \widetilde{\omega}_n \;=\; \prod_{i=1}^{k} \frac{x_i}{S(x_i)} \;=\; \prod_{i=1}^{k} (1 - S(x_i)^{p-1}).$$

(2) *Call $a \ge 0$* admissible *if $a = 0$ or the base-p digit sum of a is exactly $p-1$. For admissible $a = \sum_j \varepsilon_j p^j$, put $\varepsilon(a)! = \prod_j \varepsilon_j!$. Then*

$$\widetilde{\omega}_n \;=\; \sum_{\substack{a_1+\cdots+a_k=(p-1)n \\ a_i \ \text{admissible}}} \prod_{i=1}^{k} \frac{1}{\varepsilon(a_i)!}\, x_1^{a_1} \cdots x_k^{a_k}.$$

For $p = 2$ the admissible numbers are 0 and the powers of 2, and all coefficients are 1. This recovers the Hung–Peterson formula.

Proof. (1) Apply Λ'_t with $t = (-1, 0, 0, \dots)$. Then $T(x) = x - x^p$, so $\Lambda'_t(x) = T^{-1}(x) = S(x)$ by (3). Hence $\sum_n (-1)^n \chi(P^n)(e^{-1}) = \Lambda'_t(e^{-1}) = \prod_i S(x_i)^{-1}$. Multiply by e and take the homogeneous component of degree $(p-1)n|x_1|$. The second equality is $z/S(z) = (S(z) - S(z)^p)/S(z) = 1 - S(z)^{p-1}$.

(2) Expand $S(z)^{p-1} = \sum \binom{p-1}{\varepsilon_0,\varepsilon_1,\dots} z^{\sum \varepsilon_j p^j}$ over $\sum \varepsilon_j = p-1$. Different ε give different exponents, since there are no carries. By Wilson's theorem $-\binom{p-1}{\varepsilon} = -(p-1)!/\varepsilon! = 1/\varepsilon!$ in $\mathbb{F}_p$. □

Example 3.3. For $p = 3$, $k = 2$ we get $\widetilde{\omega}_1 = -(x_1^2 + x_2^2)$ and $\widetilde{\omega}_3 = -(x_1^6 + x_1^4 x_2^2 + x_1^2 x_2^4 + x_2^6) = c_{2,1}$. Also $\widetilde{\omega}_4 = x_1^6 x_2^2 + x_1^4 x_2^4 + x_1^2 x_2^6 = c_{2,0}$. The element $\widetilde{\omega}_1$ is not invariant. In general $\widetilde{\omega}_n$ need not be GL_k-invariant. For $p = 2$, $k = 3$, the non-invariant ones with $n \le 40$ are $n = 1, 2, 3, 5, 9, 17, 33$. For $p = 3$, $k = 2$ they are $n = 1, 2, 5, 14$.

Theorem 3.4 (Dickson invariants are Peterson elements)**.** *For $0 \le i \le k$,*

$$c_{k,i} = \widetilde{\omega}_{(p^k - p^i)/(p-1)}.$$

Proof. Let $F(X) = \sum_{i=0}^{k} \widetilde{\omega}_{(p^k-p^i)/(p-1)} X^{p^i}$. It is additive with $\mathbb{F}_p$-coefficients and monic of degree p^k, because $\widetilde{\omega}_0 = 1$. By (D1), (D2) and symmetry it suffices to show $F(x_1) = 0$. Introduce a grading variable w and put $\widetilde{\omega}(w) = \sum_n \widetilde{\omega}_n w^{(p-1)n} = \prod_l (1 - S(x_l w)^{p-1})$. The coefficient of w^{p^k} in $\widetilde{\omega}(w) S(x_1 w) = \sum_{i \geq 0} \widetilde{\omega}(w) (x_1 w)^{p^i}$ is $\sum_{i \leq k} \widetilde{\omega}_{(p^k-p^i)/(p-1)} x_1^{p^i} = F(x_1)$. By (3),

$$\widetilde{\omega}(w)\, S(x_1 w) = \bigl(S(x_1 w) - S(x_1 w)^p\bigr) \prod_{l \geq 2} (1 - S(x_l w)^{p-1}) = x_1 w \prod_{l=2}^{k} (1 - S(x_l w)^{p-1}).$$

Its w^{p^k}-coefficient is x_1 times a sum of monomials $x_2^{a_2} \cdots x_k^{a_k}$ with admissible a_l and $\sum a_l = p^k - 1$. The digit sum is subadditive, so the digit sum of $\sum a_l$ is at most $(k-1)(p-1)$. But $p^k - 1$ has digit sum $k(p-1)$. So there are no such monomials, and $F(x_1) = 0$. □

Remark 3.5. For $p = 2$, Theorem 3.4 is the formula of [5, §2]. In the code, generator number j of `DicksonAlgebra(p,k)` (counted from 1, in the order `d{deg}` is listed) is $c_{k,k-j} = \widetilde{\omega}_{n_j}$ with

$$n_j = p^{k-j} \frac{p^j - 1}{p - 1} = \frac{p^k - p^{k-j}}{p - 1}.$$

This is computed in `_left_steenrod_coaction_milnor_gen`. For example `d4,d6,d7` for $p = 2$, $k = 3$ are $\widetilde{\omega}_4, \widetilde{\omega}_6, \widetilde{\omega}_7$, and `d12,d16` for $p = 3$, $k = 2$ are $\widetilde{\omega}_3 = c_{2,1} = -Q_{2,1}$ and $\widetilde{\omega}_4 = c_{2,0} = Q_{2,0}$.

4. The algorithm

4.1. **Decomposing Peterson elements.** Let D_k^{perf} be the perfect closure of D_k, i.e. D_k with all p-power roots of its elements adjoined. Put

$$q_j = c_{k,k-j}^{\,p^{-(k-j)}} \in D_k^{\mathrm{perf}} \quad (0 \leq j \leq k), \qquad \varphi(u) = \sum_{j=0}^{k} q_j\, u^{p^j},$$

so $q_0 = 1$ and $q_k = c_{k,0}$. Since $\varphi(u) = u + O(u^p)$ is additive, it has an additive compositional inverse $\psi(w) = w + O(w^p)$.

Definition 4.1. For $n \geq 0$ and $m = (p-1)n$ let

$$\Omega_n = [u^m] \Bigl(\frac{u}{\varphi(u)}\Bigr)^m = \sum_R \binom{-m}{|R|} \binom{|R|}{r_1, r_2, \ldots, r_k} \prod_{j=1}^{k} q_j^{r_j} \ \in D_k^{\mathrm{perf}}, \tag{4}$$

where the sum runs over $R = (r_1, \ldots, r_k)$ with $\sum_j r_j (p^j - 1) = m$.

This is exactly what `PetersonPolynomials.decompose` and `DicksonAlgebra.peterson` compute. The Milnor basis enumerates the exponent vectors R. The term q^R is the Dickson monomial $\prod_j c_{k,k-j}^{\,r_j/p^{k-j}}$. When a term with nonzero coefficient has $p^{k-j} \nmid r_j$, the element lies in $D_k^{\mathrm{perf}} \setminus D_k$ and the code raises `ValueError`. Terms with $r_j \neq 0$ for some $j > k$ are dropped, which is consistent with $q_j = 0$ for $j > k$.

Proposition 4.2 (Lagrange inversion)**.** $\displaystyle\sum_{n \geq 0} \Omega_n\, w^{(p-1)n} = \frac{w}{\psi(w)}$.

Proof. Lagrange inversion in residue form holds over any commutative ring [2]. It says $[w^m]\, H(\psi(w)) = [u^m]\, H(u) \varphi'(u) (u/\varphi(u))^{m+1}$. Here $\varphi' = 1$ because φ is additive. Take $H(u) = \varphi(u)/u$. Also, $u/\varphi(u)$ is a series in u^{p-1}. □

Theorem 4.3. *$\Omega_{n_j} = c_{k,k-j}$ for $0 \leq j \leq k$. In particular the generators satisfy $\Omega_{n_j} = \widetilde{\omega}_{n_j}$.*

Proof. Put $\Omega(w) = w/\psi(w)$. As in the proof of Theorem 3.4, $\sum_j \Omega_{n_j} X^{p^{k-j}} = \left[w^{p^k}\right]\Omega(w)S(Xw)$. With $H(u) = \frac{\varphi(u)}{u}S(X\varphi(u))$, Lagrange inversion turns this into

$$\left[u^{p^k}\right]\, S(X\varphi(u))\Big(\frac{u}{\varphi(u)}\Big)^{p^k} = \left[u^{p^k}\right]\, S(X\varphi(u))\cdot\big(1+\textstyle\sum_{j\ge1} q_j^{p^k}\, u^{p^k(p^j-1)}\big)^{-1}.$$

The second factor is 1 plus terms of u-degree $\ge p^k$. The first has no constant term. So only the constant 1 contributes, and we get $\left[u^{p^k}\right]\sum_a X^{p^a}\varphi(u)^{p^a} = \sum_{a+j=k} q_j^{p^a} X^{p^a} = f_V(X)$. □

Conjecture 4.4. *If $\widetilde{\omega}_n$ is GL_k-invariant, then $\widetilde{\omega}_n = \Omega_n$.*

By Theorem 4.3 the conjecture holds at the generator degrees $n = n_j$. It has been verified in all cases listed in Table 1. The algorithm itself only needs the elements Ω_n, so it does not depend on Conjecture 4.4. The conjecture only matters for relating the code's `peterson(n)` to the polynomial $\widetilde{\omega}_n$ in other degrees.

4.2. **The action on Peterson elements.** The coefficient $a(R,n)$ of (1) is what `__action_exponents` computes for the ordinary action. The recursion `__action_exponents2` produces the multinomial $\binom{N(R)}{pr_1,p^2r_2,\dots}$ as an iterated product of binomial coefficients, and `binom_modp(p, psum+n*(p-1)-esum, psum)` is $\binom{(p-1)(n+d(R))}{N(R)}$. The operation Λ_t extends uniquely to D_k^{perf} (with values in $D_k^{\mathrm{perf}}[t^{1/p^\infty}]$). We let P^R act on D_k^{perf} by taking the coefficient of the integral monomial t^R.

Theorem 4.5 (Action formula)**.** *For all k, n and R,*

$$P^R\,\Omega_n = a(R,n)\,\Omega_{n+d(R)}.$$

This is proved in Section 5 (Theorem 5.11). The code uses a second formula for the conjugate action (`isconjugate=True`), proved in Theorem 5.14.

Theorem 4.6 (Conjugate action)**.** *For all k, n and R,*

$$\chi(P^R)\,\Omega_n = b(R,n)\,\Omega_{n+d(R)}, \qquad b(R,n) = \binom{-(p-1)n}{N(R)}\binom{N(R)}{pr_1,p^2r_2,\dots}.$$

The algorithm needs Theorem 4.5 only at the generator degrees $n = n_j$, where $\Omega_{n_j} = c_{k,k-j}$:

$$P^R c_{k,k-j} = a(R,n_j)\,\Omega_{n_j+d(R)}. \tag{5}$$

This case is proved in Theorem 5.9. If the coefficient is nonzero, then $\Omega_{n_j+d(R)} = P^R c_{k,k-j}/a(R,n_j)$ lies in D_k, so the call to `peterson` cannot fail.

Example 4.7. For $p=3$, $k=2$, $R=(0,1)$ and $n=n_1=3$ we have $d(R)=4$ and $N(R)=9$. The coefficient is $a(R,3)=\binom{14}{9}\equiv 1$, so $P(0,1)c_{2,1} = \Omega_7 = c_{2,1}c_{2,0}$. The code prints `P(0,1)*d12 = d12*d16`. For $R=(1)$ the coefficient is $\binom{8}{3}\equiv 2$, so $P^1c_{2,1} = 2c_{2,0}$, matching `P(1)*d12 = 2*d16`. In classical notation this reads $P^1Q_{2,1} = Q_{2,0}$.

5. Proofs and evidence

5.1. **A generating-function form.** Let $s_j = t_j^{1/p^j}$ and let

$$T^*(w) = \sum_{j\ge0}(t_j w)^{p^{-j}} = w + \sum_{j\ge1} s_j\, w^{p^{-j}}$$

be the adjoint of T. We work with formal sums of monomials $x^{\alpha}t^{\epsilon}w^{\beta}$ with $\epsilon \in \mathbb{Z}[1/p]^{J}_{\geq 0}$ and $\beta \in \mathbb{Z}[1/p]$, where each x-degree has only finitely many terms. These form a ring, and every expression below has only finitely many terms in each x-degree. Let π be the projection onto monomials with integral t-exponents. It is linear over elements with integral t-exponents.

Lemma 5.1. *Let $(\Omega_M)_{M\geq 0}$ be elements of degree $(p-1)M$ in a graded commutative $\mathbb{F}_p$-algebra, and $\Omega(w) = \sum_M \Omega_M w^{(p-1)M}$. Then*

$$\pi\Big[\sum_M \Omega_M\, T^*(w)^{(p-1)M}\Big] = \sum_{n\geq 0}\sum_R a(R,n)\, t^R\, \Omega_{n+d(R)}\, w^{(p-1)n}.$$

Proof. Write $T^*(w) = w(1+\sum_{j\geq 1}\vartheta_j)$ with $\vartheta_j = s_j w^{p^{-j}-1}$. Then $T^*(w)^{(p-1)M} = w^{(p-1)M}\sum_A \binom{(p-1)M}{|A|}\binom{|A|}{A}\vartheta^A$. The t_j-exponent of ϑ^A is A_j/p^j, which is integral iff $A_j = p^j r_j$. In that case $\vartheta^A = t^R w^{-(p-1)d(R)}$. Put $n = M - d(R)$. Terms with $n<0$ vanish, because then $(p-1)M < N(R) - |R| \leq N(R)$. □

On D_k^{perf} the total operation has fractional t-exponents, since $\Lambda_t(y^{1/p}) = \Lambda_t(y)^{1/p}$. For example, for $p=2$, $k=2$, $\Lambda_t(c_{2,1}^{1/2})$ contains $t_1^{1/2}c_{2,0}^{1/2}$. The operation P^R is the coefficient of the integral monomial t^R. So Theorem 4.5 says exactly that

$$\pi\big[\Lambda_t\,\Omega(w)\big] = \pi\big[\Omega(T^*w)\big], \qquad \Omega(w) = \frac{w}{\psi(w)}. \tag{6}$$

In degrees where $\Omega_n \in D_k$ (in particular for $k=1$, and at the generator degrees n_j), $\Lambda_t\Omega_n$ has integral exponents and the π on the left can be dropped. Informally: up to terms with fractional t-exponents, the total Steenrod operation acts on the Peterson series through the adjoint T^*. The mechanism is the identity

$$S\big(x\,T^*(w)\big) = S\big(T(x)\,w\big) + B(x,w), \qquad B(x,w) = \sum_{j\geq 1}\sum_{a<j} x^{p^a}(t_j w)^{p^{a-j}}, \tag{7}$$

obtained by splitting $\sum_{a,j} x^{p^a}(t_j w)^{p^{a-j}}$ according to whether $a \geq j$. Every monomial of B has t-exponent $p^{-c}\varepsilon_j$ with $c\geq 1$, where ε_j is the j-th unit vector.

Theorem 5.2. *Theorem 4.5 holds for $k=1$. Hence the algorithm of Section 4 is correct for D_1.*

Proof. For $k=1$, $\Omega_n = \widetilde{\omega}_n$, since both equal $(-1)^n\binom{-(p-1)n}{n}x^{(p-1)n}$. By Proposition 3.2, $\Omega(w) = 1 - S(xw)^{p-1}$. Therefore $\Lambda_t\Omega(w) = 1 - S(T(x)w)^{p-1}$ and $\Omega(T^*w) = 1 - S(xT^*w)^{p-1}$. By (7),

$$S(xT^*w)^e = \sum_{i=0}^{e}\binom{e}{i}S(T(x)w)^{e-i}B^i.$$

For $1\leq i\leq e\leq p-1$, every monomial of B^i has t-exponent vector equal to a sum of i vectors of the form $p^{-c}\varepsilon_j$ with $c\geq 1$. Each coordinate of such a sum is either 0 or lies in $(0,(p-1)/p]$, and at least one coordinate is nonzero, so the sum is not integral. Hence $\pi(B^i)=0$, and since $S(T(x)w)$ is integral, $\pi(S(xT^*w)^e) = S(T(x)w)^e$. So $\pi[\Omega(T^*w)] = \Lambda_t\Omega(w)$. Now compare coefficients using Lemma 5.1. □

5.2. **The general case at the generators.** For $k\geq 2$ the argument of Theorem 5.2, applied to $\prod_l(1 - S(x_l T^*w)^{p-1})$, fails. Fractional pieces coming from *different* factors can combine into integral monomials. These cross terms do not vanish in general; they are exactly why $(*)$ fails for non-invariant $\widetilde{\omega}_n$ (for example $p=2$, $k=2$, $n=1$, $R=(1)$). At the generators we can nevertheless prove the formula. The first step reduces it to a single identity.

Proposition 5.3. *Put*

$$F_t(X) = \Big[w^{p^k}\Big]\ \pi[\Omega(T^*w)]\, S(Xw) = \sum_{j=0}^{k}\Big(\sum_R a(R,n_j)\, t^R\, \Omega_{n_j+d(R)}\Big)X^{p^{k-j}}.$$

Then (5) *holds for all j and R if and only if $F_t(T(x_1)) = 0$.*

Proof. The two expressions for F_t agree by Lemma 5.1. For $j = 0$ we have $a(R,0) = \binom{N(R)-|R|}{N(R)} = 0$ unless $R = 0$, so F_t is additive and monic of degree p^k, with coefficients in $D_k^{\mathrm{perf}}[t]$. It is symmetric in the x_i. So if $F_t(T(x_1)) = 0$, then F_t vanishes on $T(V)$, and by (D2) $F_t = f_{T(V)} = \Lambda_t f_V$ by (2). Comparing coefficients gives (5). The converse is the same computation read backwards. □

The rest of this subsection proves $F_t(T(x_1)) = 0$. We need three tools: a factorisation of the Peterson series, a bound on its support, and an explicit formula for $F_t(T(X))$. Throughout, $\alpha(m)$ denotes the base-p digit sum of $m \geq 0$. Recall that $\alpha(a+b) \leq \alpha(a)+\alpha(b)$, with equality iff the addition has no carries. By Lucas' theorem, a multinomial coefficient $\binom{a_1+\cdots+a_m}{a_1,\ldots,a_m}$ is nonzero mod p iff the addition $a_1+\cdots+a_m$ has no carries. Write $g(z) = 1 - S(z)^{p-1} = z/S(z)$ and $\gamma_m = [z^m]\, g(z)$. By Proposition 3.2, $\gamma_m \neq 0$ only if $m = 0$ or $\alpha(m) = p-1$.

Factorisation. Let $0 \neq v \in V$, $f_v(X) = X^p - v^{p-1}X$ and $U = f_v(V)^{1/p}$, an $\mathbb{F}_p$-space of rank $k-1$. Put $\psi_v(w) = S(vw)/v$.

Lemma 5.4. $\Omega_V(w) = g(vw)\cdot \Omega_U(\psi_v(w))$.

Proof. By (D2), $f_V = f_{f_v(V)} \circ f_v$. Write additive series as twisted polynomials in the Frobenius F, with adjoint $(aF^i)^* = F^{-i}a$ and $(AB)^* = B^*A^*$. Then $\varphi_V = f_V^* F^k$, and

$$\varphi_V = f_v^*\, f_{f_v(V)}^*\, F^k = (f_v^* F)\circ (F^{-1}(f_{f_v(V)}^* F^{k-1})F) = \varphi_{\langle v\rangle}\circ\varphi_U,$$

because conjugating by F takes p-th roots of the coefficients. Hence $\psi_V = \psi_U \circ \psi_{\langle v\rangle}$ and $w/\psi_V(w) = \frac{w}{\psi_{\langle v\rangle}(w)}\cdot\frac{\psi_{\langle v\rangle}(w)}{\psi_U(\psi_{\langle v\rangle}(w))}$. For rank one, $\varphi_{\langle v\rangle}(u) = u - v^{p-1}u^p$, so $\psi_{\langle v\rangle} = \psi_v$ and $w/\psi_v(w) = vw/S(vw) = g(vw)$. □

Lemma 5.5 (Support)**.** *Let $\eta(w) = \sum_{a\geq 0} e_a w^{p^a}$ be any additive series. Then $\Big[w^E\Big]\ \Omega_V(\eta(w)) = 0$ unless $\alpha(E) \leq k(p-1)$.*

Proof. Induction on k; for $k = 0$, $\Omega_V = 1$. By Lemma 5.4, $\Omega_V(\eta(w)) = g(v\eta(w))\,\Omega_U(\eta'(w))$ with $\eta' = \psi_v\circ\eta$ additive. The first factor is $1 - S(v\eta(w))^{p-1}$, and $S(v\eta(w))$ is additive in w. So the first factor is supported in exponents that are 0 or a sum of $p-1$ powers of p, i.e. of digit sum $\leq p-1$. By induction the second factor is supported in digit sum $\leq (k-1)(p-1)$. Now use subadditivity of α. □

For $D_0 \geq 1$ put

$$F_t^{D_0}(X) = \Big[w^{D_0}\Big]\ \pi[\Omega(T^*w)]\, S(Xw),$$

so that $F_t = F_t^{p^k}$. This is an additive polynomial in X.

Lemma 5.6. *Fix $D_0 \geq 1$. For R put $D_R = D_0 + (p-1)d(R)$, and write $\Big[y^{pR}\Big]$ for the coefficient of $\prod_j y_j^{p^j r_j}$. Put*

$$b_R = \Big[y^{pR}\Big]\,(1+\textstyle\sum_j y_j)^{D_R}, \qquad \delta_{R,e} = \Big[y^{pR}\Big]\,\Big(\sum_{j>e} y_j^{p^e}\Big)(1+\textstyle\sum_j y_j)^{D_R - p^e}.$$

Then

$$F_t^{D_0}(T(X)) = \sum_R t^R \left[w^{D_R}\right]\, \Omega(w)\, B_R(Xw), \qquad B_R(z) = b_R\, S(z) - \sum_{e\geq 0} \delta_{R,e}\, z^{p^e}.$$

Proof. Since $S(T(X)w)$ has integral t-exponents, $F_t^{D_0}(T(X)) = \pi\left[w^{D_0}\right]\Omega(T^*w)\,S(T(X)w)$. By (7), $S(T(X)w) = S(XT^*w) - B(X,w)$. The first part is $\pi\left[w^{D_0}\right]H(T^*w)$ with $H(w) = \Omega(w)S(Xw)$. Expanding as in Lemma 5.1 gives $\sum_R t^R b_R\left[w^{D_R}\right]H(w)$. For the second part, take a monomial $X^{p^a}(t_j w)^{p^{a-j}}$ of B (so $a<j$) times the term $\Omega_M w^{(p-1)M}\binom{(p-1)M}{|A|}\binom{|A|}{A}\vartheta^A$ of $\Omega(T^*w)$. Its t-exponents are integral iff $A_i = p^i r_i$ for $i\neq j$ and $A_j = p^j r_j - p^a$ with $r_j\geq 1$. Then its w-exponent is $(p-1)M - (p-1)d(R) + p^a$, which equals D_0 iff $(p-1)M = D_R - p^a$. Summing the coefficients over $j>a$ gives $\delta_{R,a}$. So the second part is $-\sum_R t^R \sum_a \delta_{R,a} X^{p^a}\Omega_{(D_R-p^a)/(p-1)} = -\sum_R t^R\left[w^{D_R}\right]\Omega(w)\sum_a \delta_{R,a}(Xw)^{p^a}$. □

Lemma 5.7 (Digit sums)**.** *Let $D_0\geq 1$, R arbitrary, $e\geq 0$ and $D = D_R = D_0 + (p-1)d(R)$.*

1. *If $\left[y^{pR}\right](1+\sum_j y_j)^{D-1}\neq 0$, then $\alpha(D-1)\geq \alpha(D_0-1)$.*
2. *If $\delta_{R,e}\neq 0$, then $\alpha(D-p^e)\geq \alpha(D_0-1)+p-1$.*

Proof. Recall $D-N(R) = D_0-|R|$. (1) The coefficient is $\binom{D-1}{N(R)}\binom{N(R)}{pR}$. If it is nonzero, the parts $p^j r_j$ add without carries, so $\alpha(N(R)) = \sum_j \alpha(r_j)\geq \alpha(|R|)$. Moreover $D-1-N(R) = D_0-1-|R|\geq 0$ and $N(R)+(D_0-1-|R|)$ has no carries. Since $\alpha(D_0-1)\leq \alpha(D_0-1-|R|)+\alpha(|R|)$, we get $\alpha(D-1) = \alpha(N(R))+\alpha(D_0-1-|R|)\geq \alpha(D_0-1)$.

(2) Some summand $\binom{D-p^e}{N(R)-p^e}\binom{N(R)-p^e}{pR-p^e\varepsilon_j}$ with $j>e$ and $r_j\geq 1$ is nonzero. The parts $p^i r_i$ $(i\neq j)$ and $p^j r_j - p^e = p^e(p^{j-e}(r_j-1)+p^{j-e}-1)$ add without carries, so

$$\alpha(N(R)-p^e) = \sum_{i\neq j}\alpha(r_i) + \alpha(r_j-1) + (j-e)(p-1).$$

The binomial coefficient is nonzero, so $D-N(R) = D_0-|R|\geq 0$ (with $|R|\geq 1$) and $\alpha(D-p^e) = \alpha(N(R)-p^e)+\alpha(D_0-|R|)$. Moreover $\alpha(D_0-|R|)\geq \alpha(D_0-1)-\alpha(|R|-1)$ and $\alpha(|R|-1)\leq \sum_{i\neq j}\alpha(r_i)+\alpha(r_j-1)$. Hence

$$\alpha(D-p^e)\geq (j-e)(p-1)+\alpha(D_0-1)\geq \alpha(D_0-1)+p-1.$$ □

Proposition 5.8 (Vanishing)**.** *If $\alpha(D_0-1) > (k-1)(p-1)$, then $F_t^{D_0}(T(v)) = 0$ for all $v\in V$.*

Proof. By additivity it suffices to take $v\neq 0$. By Lemma 5.4, $\Omega_V(w) = g(vw)G(w)$ with $G(w) = \Omega_U(\psi_v(w))$. Since $g(z)S(z) = z$,

$$g(z)B_R(z) = \sum_E G_{R,E}\, z^E, \qquad G_{R,E} = b_R\,[E=1] - \sum_{e\geq 0}\delta_{R,e}\,\gamma_{E-p^e}.$$

By Lemma 5.6, the coefficient of t^R in $F_t^{D_0}(T(v))$ is $\sum_E G_{R,E}\, v^E\left[w^{D_R-E}\right]G(w)$. We show that every term vanishes. Suppose $\left[w^{D_R-E}\right]G\neq 0$. By Lemma 5.5 (rank $k-1$, $\eta=\psi_v$), $\alpha(D_R-E)\leq (k-1)(p-1)$.

- If $E=1$, then $G_{R,1} = b_R - \delta_{R,0} = \left[y^{pR}\right](1+\sum y_j)^{D_R-1}$. If this were nonzero, Lemma 5.7(1) would give $\alpha(D_R-1)\geq \alpha(D_0-1) > (k-1)(p-1)$. This is a contradiction.

- If $E > 1$, a nonzero summand $\delta_{R,e}\gamma_{E-p^e}$ has $\alpha(E-p^e) \le p-1$. Hence $\alpha(D_R - p^e) \le \alpha(D_R - E) + \alpha(E-p^e) \le k(p-1)$. But Lemma 5.7(2) gives $\alpha(D_R - p^e) \ge \alpha(D_0 - 1) + p - 1 > k(p-1)$.

So $G_{R,E} = 0$ whenever $\left[w^{D_R - E}\right] G \neq 0$, and $F_t^{D_0}(T(v)) = 0$. □

Theorem 5.9 (Action on the generators)**.** *Formula* (5) *holds for all k, j and R:*

$$P^R c_{k,k-j} = a(R, n_j)\, \Omega_{n_j + d(R)}.$$

Proof. Take $D_0 = p^k$, for which $\alpha(p^k - 1) = k(p-1) > (k-1)(p-1)$. Proposition 5.8 gives $F_t(T(x_1)) = 0$; now apply Proposition 5.3. □

Corollary 5.10. *The algorithm of Section 4 computes the correct $\mathcal{A}$-action on D_k for all p and k. Moreover, if $a(R, n_j) \neq 0$, then $\widetilde{\omega}_{n_j + d(R)} = \Omega_{n_j + d(R)}$. So Conjecture 4.4 holds in all degrees that the algorithm uses.*

Proof. The first claim follows from Theorem 5.9 and Section 6. For the second, run the same proof with the polynomial family $\widetilde{\omega}_n$ in place of Ω_n. Lemmas 5.1 and 5.6 hold for any family. The factorisation is $\widetilde{\omega}(w) = g(x_1 w) \prod_{l \ge 2} g(x_l w)$ (Proposition 3.2), and the second factor is supported in digit sum $\le (k-1)(p-1)$. So $P^R c_{k,k-j} = a(R, n_j)\, \widetilde{\omega}_{n_j + d(R)}$ as well. □

For single powers P^i, Theorem 5.9 recovers the known formulas of Hung [3] ($p = 2$) and Hung–Minh [4] (odd p; see also [10, Thm. 3.4]). We compared these with the code's output for all s and i for $p = 2$, $k = 3, 4$; $p = 3$, $k = 2, 3$; and $p = 5, 7$, $k = 2$, using $c_{k,s} = (-1)^{k-s} Q_{k,s}$. All 298 values agree.

5.3. **All degrees.**

Theorem 5.11. *Theorem 4.5 holds: $P^R \Omega_n = a(R, n)\, \Omega_{n + d(R)}$ for all k, n and R.*

Proof. Put $G = \pi[\Omega_V(T^* w)]$ and $G' = \pi[\Lambda_t \Omega_V(w)] = \pi[\Omega_{T(V)}(w)]$. By Lemma 5.1 we must show $G = G'$. Let $\Delta = G' - G = \sum_n \Delta_n w^{(p-1)n}$. Then $\Delta_0 = 0$, and every Δ_n has integral t-exponents.

Step 1: a family of linear conditions. Let $\alpha(D_0 - 1) > (k-1)(p-1)$ and $u \in T(V)$. By Proposition 5.8, $\left[w^{D_0}\right] G(w) S(uw) = F_t^{D_0}(u) = 0$. Lemmas 5.4 and 5.5 hold for the rank-k space $T(V)$ as well. So $\Omega_{T(V)}(w) S(uw) = uw\, \Omega_{U'}(\psi_u(w))$ is supported in exponents E with $\alpha(E-1) \le (k-1)(p-1)$. Since $S(uw)$ is integral, the same holds for $G'(w) S(uw)$. Hence the additive polynomial

$$L_{D_0}(X) = \left[w^{D_0}\right] \Delta(w) S(Xw) = \sum_{e \ge 0} \Delta_{(D_0 - p^e)/(p-1)} X^{p^e}$$

vanishes on $T(V)$. Divide L_{D_0} by the monic additive polynomial $f_{T(V)}$: $L_{D_0} = M \circ f_{T(V)} + \mathrm{Rem}$, where Rem has degree $< p^k$. The usual division algorithm works for additive polynomials over any ring of characteristic p. Since Rem vanishes on the p^k elements of $T(V)$, $\mathrm{Rem} = 0$.

Step 2: a recursion. Let $C = \{n \ge 1 : p^{k-1} \mid (p-1)n\}$. For $n \in C$ put $D_0 = (p-1)n + p^{k-1}$. The low $k-1$ digits of $(p-1)n$ vanish, so $\alpha(D_0 - 1) = \alpha((p-1)n) + (k-1)(p-1) > (k-1)(p-1)$. In L_{D_0}:

- the term $e = k-1$ is $\Delta_n X^{p^{k-1}}$;
- the terms $e < k-1$ are X^{p^e} with $e < k-1$, and are already reduced;
- the terms $e \ge k$ have indices $n_e = n - (p^e - p^{k-1})/(p-1) < n$. Here $(p-1)n_e \equiv 0 \bmod p^{k-1}$, so $n_e \in C \cup \{0\}$, or the term is absent.

Reducing X^{p^e} ($e \geq k$) with $X^{p^k} \equiv -\sum_{i<k} c_i(T(V))X^{p^i}$, the coefficient of $X^{p^{k-1}}$ in Rem is $\Delta_n + \sum_{e\geq k} \rho_e \Delta_{n_e}$ for certain ρ_e. It vanishes by Step 1. By induction on $n \in C$, starting from $\Delta_0 = 0$, we get $\Delta_n = 0$ for all $n \in C$.

Step 3: integral degrees. Let $N = p^k m$ with $m \geq 1$; then $N \in C$. Moreover $\Omega_N = \Omega_m^{p^k}$: raise $(u/\varphi)^{(p-1)m}$ to the p^k-th power in (4). This lies in D_k, because $q_j^{p^k} \in D_k$. So $\Lambda_t\Omega_N$ has integral t-exponents, and Step 2 gives

$$\Lambda_t\Omega_N = \sum_S t^S\, a(S,N)\, \Omega_{N+d(S)}.$$

Step 4: descent. For arbitrary m, $\Lambda_t\Omega_m = (\Lambda_t\Omega_{p^k m})^{p^{-k}}$. The coefficient of the integral monomial t^R comes from $S = p^k R$. It is $\big(a(p^kR, p^km)\,\Omega_{p^k(m+d(R))}\big)^{p^{-k}} = a(R,m)\,\Omega_{m+d(R)}$, by Lucas' theorem and $\Omega_{p^k j} = \Omega_j^{p^k}$. □

For $k = 1$, C consists of all $n \geq 1$, and Step 2 is Theorem 5.2 again.

5.4. **The conjugate action.** On D_k^{perf} we let $\chi(P^R)$ act by the integral coefficients of $\Lambda_t' = \Lambda_t^{-1}$, as for P^R.

The proof reduces the conjugate formula to the ordinary one, using the antipode identity and a vanishing statement for binomial coefficients.

Proposition 5.12. *Suppose that, for all $n \geq 0$ and $R \neq 0$,*

$$\sum_{R'+R''=R} b(R',\, n + d(R''))\; a(R'',\, n) = 0. \tag{8}$$

Then Theorem 4.6 holds.

Proof. By the same descent as in Step 4, it suffices to treat N with $\Omega_N \in D_k$. The element $\chi(P^R)$ is a linear combination of Milnor basis elements P^S with $d(S) = d(R)$. So by Theorem 5.11, $\chi(P^R)\Omega_N$ is a scalar multiple of $\Omega_{N+d(R)}$. We show by induction on $|R|$ that the scalar is $b(R,N)$, for all such N. Apply $\sum_{R'+R''=R} \chi(P^{R'})P^{R''} = 0$ (the antipode identity, $R \neq 0$) to Ω_N. If $a(R'',N) \neq 0$, then $\Omega_{N+d(R'')} = P^{R''}\Omega_N / a(R'',N)$ lies in D_k, so for $R'' \neq 0$ the induction hypothesis applies to it. Thus

$$\chi(P^R)\Omega_N = -\sum_{R''\neq 0} b(R', N + d(R''))\, a(R'', N)\, \Omega_{N+d(R)} = b(R,N)\,\Omega_{N+d(R)}$$

by (8), since $a(0,\cdot) = 1$. □

To prove (8) we need a projection on power series. Let $y_1, \ldots, y_J$ be variables, $Y = 1 + \sum_i y_i$, and fix levels $\lambda = (\lambda_1, \ldots, \lambda_J) \in \mathbb{Z}_{\geq 0}^J$. Let Π_λ keep the monomials y^E with $p^{\lambda_i} \mid E_i$ for all i, and put $\|E\|_\lambda = \sum_i E_i/p^{\lambda_i}$ for such E. For $m \in \mathbb{Z}$, Y^m is a power series, a polynomial if $m \geq 0$.

Lemma 5.13. *Let $E \neq 0$ with $p^{\lambda_i} \mid E_i$, and let $\mu, \nu \in \mathbb{Z}$ with $\mu + \nu = |E| - \delta$, where $1 \leq \delta \leq \|E\|_\lambda$. Then*

$$\big[y^E\big]\; \Pi_\lambda(Y^\mu)\, \Pi_\lambda(Y^\nu) = 0.$$

Proof. Write $m = \sum_a m_a p^a$ in base p (p-adically if $m < 0$). By Lucas, $Y^m = \prod_a (1 + \sum_i y_i^{p^a})^{m_a}$. In this product the exponent of y_i is $\sum_a p^a k_{a,i}$ with digits $k_{a,i} \leq m_a < p$, so it is divisible by

p^{λ_i} iff $k_{a,i} = 0$ for $a < \lambda_i$. Hence

$$\Pi_\lambda(Y^m) = \prod_{a\ge 0}\Big(1 + \sum_{i:\lambda_i\le a} y_i^{p^a}\Big)^{m_a} = \Big(1+\sum_{i\in Z} y_i\Big)^{m_0}\cdot \Pi_{\lambda^-}(Y^{\lfloor m/p\rfloor})^p,$$

where $Z = \{i : \lambda_i = 0\}$, $\lambda_i^- = \max(\lambda_i - 1, 0)$ and $\lfloor m/p\rfloor = (m - m_0)/p$. We use induction on $\sum_{i\notin Z,\, E_i\neq 0}\lambda_i$.

If $E_i = 0$ for all $i \notin Z$, set those $y_i = 0$. Then $\Pi_\lambda(Y^m)$ becomes $(1+\sum_{i\in Z} y_i)^m$, and the coefficient is $\binom{|E|-\delta}{|E|}\binom{|E|}{E} = 0$, since $0 \le |E| - \delta < |E|$.

Otherwise let $s = \mu_0 + \nu_0 \le 2p-2$, $\mu' = \lfloor \mu/p\rfloor$, $\nu' = \lfloor \nu/p\rfloor$. Then the coefficient is a sum, over splittings $E_i = e_i + pe_i'$ ($i \in Z$), of $[y^e]\,(1+\sum_{i\in Z} y_i)^s$ times $\left[y^{E''}\right]\Pi_{\lambda^-}(Y^{\mu'})\Pi_{\lambda^-}(Y^{\nu'})$. Here $E_i'' = E_i/p$ for $i \notin Z$ and $E_i'' = e_i'$ for $i \in Z$. Only splittings with $|e| \le s$ contribute. We check the hypotheses for $(\lambda^-, E'', \mu', \nu')$. We have $|E| = p|E''| + |e|$, so $\mu'+\nu' = (\mu+\nu-s)/p = |E''| - \delta''$ with $\delta'' = (\delta + s - |e|)/p$, an integer. Since $\delta \ge 1$ and $|e| \le s$, $\delta'' \ge 1$. Let $c = \sum_{i\notin Z} E_i/p^{\lambda_i} \ge 1$, so that $\|E\|_\lambda = c + |e| + p|e'|$ and $\|E''\|_{\lambda^-} = c + |e'|$. Then $\delta'' \le (c+s)/p + |e'| < c + 1 + |e'|$, using $s \le 2p-2 < (p-1)c + p$. So $\delta'' \le \|E''\|_{\lambda^-}$. If $E'' = 0$, then $|E| = |e| \le s$ and $c = 0$, which is excluded. So the induction hypothesis applies, and every term vanishes. □

Theorem 5.14. *Identity* (8) *holds. Hence Theorem 4.6 holds.*

Proof. Let $\rho_a(t)$ and $\rho_b(t)$ be the operators on formal symbols $(E_n)_{n\ge 0}$ given by $E_n \mapsto \sum_R t^R a(R,n) E_{n+d(R)}$ and $E_n \mapsto \sum_R t^R b(R,n) E_{n+d(R)}$. Identity (8) says $\rho_b\rho_a = 1$. Both operators are the identity plus terms of positive t-degree, so this is equivalent to $\rho_a\rho_b = 1$, i.e. to

$$\sum_{R_1+R_2=R} b(R_1,n)\, a(R_2, n+d(R_1)) = 0 \qquad (R \neq 0).$$

Now $b(R_1,n) = \left[y^{pR_1}\right] Y^{-(p-1)n}$, where $y^{pR} = \prod_j y_j^{p^j r_j}$. Similarly $a(R_2, n+d(R_1)) = \left[y^{pR_2}\right] Y^{(p-1)n + N(R) - |R|}$, because $(p-1)(n + d(R_1) + d(R_2)) = (p-1)n + N(R) - |R|$. So the sum is $\left[y^{pR}\right]\Pi_\lambda(Y^\mu)\Pi_\lambda(Y^\nu)$ with $\lambda_j = j$, $\mu = -(p-1)n$, $\nu = (p-1)n + N(R) - |R|$. Here $E = pR$ has $|E| = N(R)$ and $\|E\|_\lambda = |R|$, and $\mu + \nu = |E| - |R|$. Lemma 5.13 with $\delta = |R|$ shows that it vanishes. Now apply Proposition 5.12. □

5.5. **Computer verification.** All checks below compute the action directly on polynomials in $\mathbb{F}_p[x_1, \dots, x_k]$, by substituting $x_i \mapsto T(x_i)$ (or $x_i \mapsto T^{-1}(x_i)$ for χ) and extracting coefficients. They are independent of yacop's implementation. Every check that was run succeeded.

The module doctests also check associativity $(ab)m = a(bm)$ of the action the code defines (`_manual_test_left_action`).

6. The action on the whole algebra

The code computes $P^R(m)$ for a monomial m through a *coaction*: a list of triples (y, Q, R'), standing for $y \otimes \xi^{R'}$ (and exterior τ^Q). Then $P^R(m)$ is the sum of the y with $R' = R$ (`_act_from_coact` in `algebra_base.py`). The ingredients are:

(1) *Generators.* For $c_{k,k-j}$ the coaction is $\sum_R a(R,n_j)\,\Omega_{n_j+d(R)} \otimes \xi^R$, truncated to the R that are dominated by the requested operation.

(2) *Powers.* The function `_factor_key` writes an exponent in base p, $g^e = \prod_u (g^{p^u})^{\varepsilon_u}$ with $0 \le \varepsilon_u < p$. The coaction of g^{p^u} is that of g with everything raised to the p^u-th power, since $P^R(y^{p^u}) = (P^{R/p^u} y)^{p^u}$, or 0 if $p^u \nmid R$. For $\varepsilon_u > 1$ a multinomial expansion is used (`_multinomials`).

statement	range	cases
$(*)$ for $\widetilde{\omega}_n$, $k=1$ (exact formula)	$p=3,5,7$, $n<60$	1350
$(*)$, invariant $\widetilde{\omega}_n$	$p=2$: $k=2,3$; $p=3$: $k\le 3$; $p=5,7$: $k=2$	682
conjugate formula, invariant $\widetilde{\omega}_n$	$p=2$: $k=2,3$; $p=3$: $k\le 3$; $p=5$: $k\le 2$	321
Conjecture 4.4	$p=2$: $k=3,4$, $n\le 40$; $p=3$: $k=2$, $n\le 40$; $k=3$, $n\le 30$; $p=5$: $n\le 40$; $p=7$: $n\le 30$ $(k=2)$	143
$(*)$ for Ω_n, all n (in D_k^{perf})	$p=2$: $k=2,3$; $p=3$: $k=2$	134
$(*)$ for $P^{\Delta_j}=P(0,\dots,0,1)$, invariant $\widetilde{\omega}_n$	$p=2$: $k=3$ $(j=2,3)$, $k=4$ $(j=2)$; $p=3$: $k=3$, $j=2$; $p=5$: $k=2$, $j=2$	75
$G_{R,E}=0$ when $\alpha(D_R-E)\le(k-1)(p-1)$	$p=2,3,5$; $k\le 4$; $\ell(R)\le 3$, $r_j\le p+1$	$\approx 10^6$
same, general D_0 (Prop. 5.8)	$p=2,3,5$; $k\le 3$; $\ell(R)\le 2$; $D_0<p^{k+1}+p$	$1.3\cdot 10^6$
identity (8)	$p=2,3$: $\ell(R)\le 3$; $p=5$: $\ell(R)\le 2$; $n<25$	7175
Lemma 5.13	$p=2,3$; $J\le 3$; $\lambda_i\le 2$; $\lvert\mu\rvert\le 25$	240516

TABLE 1. Computer checks of the action formula. "Cases" counts pairs (n,R) (or values of n for the decomposition), with R ranging over small Milnor indices.

(3) *Products.* Coactions of factors are multiplied (`_coaction_helper`). This is the Cartan formula $P^R(yz)=\sum_{R'+R''=R}P^{R'}(y)P^{R''}(z)$, i.e. multiplicativity of Λ_t.
(4) *Negative powers.* In the localised algebra (the elements `_zetapower`, `_xipower` involve $c_{k,0}^{-1}$), a negative exponent $-e$ is written as $-p^u+(p^u-e)$, with p^u larger than every entry of R. Then $c_{k,0}^{-p^u}$ is invariant under all relevant $P^{R'}$.
(5) *Mùi generators.* The exterior generators τ_n are handled by the coaction of Theorem 7.1 (`_tau_coaction`), using the Dickson part for the coefficients $\Lambda(\rho_{N,j})$; see Section 7.

Because D_k is a polynomial algebra, (1)–(3) determine the action completely. Together with Theorem 5.9 this proves that the D_k part of the code is correct for the ordinary action. The conjugate action (`left_steenrod_action_milnor_conj`) uses Theorem 4.6 on the Dickson generators, so its D_k part is correct too. The code also offers an independent route, `admissible_action`, through admissible matrices, and restriction maps $D_k\to D_{k'}$ for $k'<k$; these are not discussed here.

7. The Mùi generators

Let p be odd, or $p=2$ with the generic Steenrod algebra. Then $H^*(BV_k)=E(y_1,\dots,y_k)\otimes H$ with $|y_i|=1$ and $\beta y_i=x_i$. We extend the total operation by odd parameters $e_0,e_1,\dots$ dual to the Milnor primitives Q_l:

$$\Lambda(x)=T(x)=\sum_j t_j x^{p^j},\qquad \Lambda(y)=y+E(x),\quad E(X)=\sum_l e_l X^{p^l}.$$

This is Milnor's coaction $\lambda(y)=y\otimes 1+\sum_l x^{p^l}\otimes\tau_l$ in total form.

The generators. The Moore matrix $(x_i^{p^j})_{i,j<k}$ is invertible over $\mathrm{Frac}(H)$. So there are unique elements $\tau_0,\dots,\tau_{k-1}$, linear in the y_i, with

$$y_v=\tau(x_v):=\sum_{j<k}\tau_j\,x_v^{p^j}\qquad\text{for all } v\in V, \tag{9}$$

where y_v,x_v denote the classes corresponding to v. By uniqueness the τ_j are GL_k-invariant, and $|\tau_j|=1-2p^j$. Up to sign, $\tau_j=M_{k,j}/L_k$ is Mùi's invariant divided by the Moore determinant, equivalently $R_{k,j}/Q_{k,0}$ [8]. These are the generators `tau{j}` of `DicksonAlgebra(p,k)`

(with `generic=True` if $p = 2$). Write $\rho_{N,j} \in D_k$ for the coefficients of the reduction $X^{p^N} \equiv \sum_{j<k} \rho_{N,j} X^{p^j} \bmod f_V$ (`_reduced_frobenius`). Also let $C = T^{-1} = \sum_m c_m X^{p^m}$, so that $c_m = \chi(\xi_m)(t)$.

Theorem 7.1.

$$\Lambda(\tau_j) = \sum_{N\geq 0} \Big[\sum_{i+m=N} \tau_i \, c_m^{p^i} + \sum_{l+m=N} e_l \, c_m^{p^l} \Big] \Lambda(\rho_{N,j}).$$

In particular $Q_l \tau_j = \rho_{l,j}$, which is δ_{lj} for $l < k$.

Proof. Apply the ring homomorphism Λ to (9). This gives $\Lambda(\tau)(T(v)) = y_v + E(x_v) = (\tau + E)(v)$ for $v \in V$, where $\Lambda(\tau)$ is the additive polynomial $\sum_j \Lambda(\tau_j) X^{p^j}$. Consider the additive series $(\tau + E) \circ C$, which converges t-adically. It takes the same values on $T(V)$ as $\Lambda(\tau)$. Divide by the monic polynomial $f_{T(V)} = \Lambda(f_V)$. The remainder of $\Lambda(\tau) - (\tau + E) \circ C$ has degree $< p^k$ and vanishes on the p^k elements of $T(V)$, so it is zero. Since $\Lambda(\tau)$ already has degree $< p^k$, it equals the remainder of $(\tau + E) \circ C$. Now $(\tau + E) \circ C = \sum_N [\sum_{i+m=N} \tau_i c_m^{p^i} + \sum_{l+m=N} e_l c_m^{p^l}] X^{p^N}$. The remainder of X^{p^N} modulo $f_{T(V)}$ is $\sum_j \Lambda(\rho_{N,j}) X^{p^j}$, because $\rho_{N,j}$ is a polynomial in the Dickson invariants. □

In coaction language, $t_j \leftrightarrow \xi_j$, $c_m \leftrightarrow \chi(\xi_m)$ and $e_l \leftrightarrow \tau_l$. The coefficients $\Lambda(\rho_{N,j})$ are computed by the Dickson part (Theorem 5.11). The conjugate coaction is obtained by applying $1 \otimes \chi$. In it $c_m \leftrightarrow \xi_m$ and $e_l \leftrightarrow \chi(\tau_l) = -\sum_{j \leq l} \chi(\xi_{l-j})^{p^j} \tau_j$.

Remark 7.2. Earlier versions of the code used the model $\Delta\tau_n = \tau_n \otimes 1 + \sum_i \xi_{n-i}^{p^i} \otimes \tau_i$, with ξ_j the elements `_xipower(j)` of $D_k[c_{k,0}^{-1}]$. It made reduced powers kill the τ_n. This is not an $\mathcal{A}$-module. For $p = 2$, $k = 3$, a direct polynomial computation gives $P(8)(c_{3,1}/c_{3,0}) = c_{3,0} \neq 0$, whereas $P(8)\xi_1 = 0$ in $\mathcal{A}_*$. Consequently $(P(8)Q_0)(\tau_1\tau_2) = 0 \neq P(8)(Q_0(\tau_1\tau_2))$ in that model. By Theorem 7.1, reduced powers do not kill the τ_n; for example $P^1\tau_1 = -\tau_0$.

Verification. For $p = 3$ ($k = 1, 2$), $p = 5$ ($k = 2$) and $p = 2$ generic ($k = 2$), the new code agrees with a direct computation of $P^r\tau_j$ in $H^*(BV_k)$, where τ_j is written as an explicit linear form in the y_i with rational coefficients. The action passes the associativity tests, for both the ordinary and the conjugate action, on the regions where the old model failed.

8. The dual Dickson algebras

The class `DicksonDualSteenrodAlgebra(p,k)` implements

$$\mathcal{A}_*(k) = \mathbb{F}_p[\xi_1, \ldots, \xi_k] \otimes E(\tau_0, \ldots, \tau_{k-1}) \subset \mathcal{A}_*,$$

with the generators placed in negative degrees ($|\xi_j| = -2(p^j - 1)$, $|\tau_j| = 1 - 2p^j$). The code prints the ξ_j as `zeta[j]`. Milnor's coproduct $\Delta\xi_n = \sum_i \xi_{n-i}^{p^i} \otimes \xi_i$ only involves indices $\leq n$, and similarly for τ_n. So $\mathcal{A}_*(k)$ is a sub-bialgebra of $\mathcal{A}_*$. It is an $\mathcal{A}$-module algebra via $\theta \cdot y = \sum y_{(1)} \langle \theta, y_{(2)} \rangle$ (`_coproduct_gen` and `_left_steenrod_coaction_milnor_gen`).

It is related to the Dickson algebras in two ways.

(a) The localised Dickson algebra. The algebra map $\xi_i \mapsto c_{k,i}/c_{k,0}$ (`DicksonAlgebra._zetapower`) sends $\mathcal{A}_*(k)$'s polynomial part into $D_k[c_{k,0}^{-1}]$ and preserves degrees. The code uses its conjugate (`_xipower`) for the Mùi generators, as explained in Section 6.

(b) Peterson elements are images of a canonical element of $\mathcal{A}_*$. Let $\xi(w) = \sum_j \xi_j w^{p^j}$ and $\zeta(w) = \sum_j \chi(\xi_j) w^{p^j}$. These are compositional inverses of each other, because $\sum_i \chi(\xi_i) \xi_{n-i}^{p^i} = 0$

for $n \geq 1$. Let $\mu\colon \mathcal{A}_* \to D_k^{\mathrm{perf}}$ be the algebra map with $\chi(\xi_j) \mapsto q_j$ for $j \leq k$ and $\chi(\xi_j) \mapsto 0$ for $j > k$. Then $\mu(\zeta) = \varphi$, $\mu(\xi) = \psi$, and by Proposition 4.2

$$\sum_n \Omega_n\, w^{(p-1)n} = \mu\Big(\frac{w}{\xi(w)}\Big) = \mu\Big(\sum_R (-1)^{|R|}\binom{|R|}{R}\xi^R\, w^{(p-1)d(R)}\Big).$$

The roots of φ form an $\mathbb{F}_p$-subspace Y of dimension k, in degree $-|x_1|$, of an algebraic closure. Since φ has linear coefficient 1 and top coefficient $c_{k,0}$, we have $\varphi = f_Y/c_0(Y)$ and $q_j = c_j(Y)/c_0(Y)$. So, up to the conjugation χ, μ is the map of (a) for the "dual" space Y in place of V. In this sense the Peterson elements of V are the images of the canonical elements $\sum_{d(R)=n}(-1)^{|R|}\binom{|R|}{R}\xi^R \in \mathcal{A}_*(k)$ under the Dickson map of the dual space. Equation (6), with the adjoint T^*, suggests that this duality is the natural setting for these formulas.

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